ModelIt!

Student Activity Pack

Blood Moon: When the Sky Turns Red

The Cosmic Dance of Earth, Moon, and Sun

Grade 7 | NGSS MS-ESS1-1, MS-ESS1-2 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think the Moon appears to change shape throughout the month?

2 Have you ever seen a lunar eclipse? What did it look like?

3 Draw what you think causes the Moon to look different each night.

4 Does the Moon make its own light? How do we see it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sun Angle | Moon Position | Earth's Shadow | Visible Moon Phase

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Moon

Moon on opposite side of Earth from Sun, not in shadow

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Lunar Eclipse

Moon passes through Earth's Shadow directly

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: New Moon

Moon between Earth and Sun

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tidal Force — The gravitational pull of the Moon that creates ocean tides on Earth

Orbital Tilt — The 5-degree tilt of the Moon's orbit that determines when eclipses can occur

Solar Eclipse — When the Moon passes between Earth and Sun, blocking sunlight

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does the Moon change shape every night and sometimes turn blood red? Use evidence from your simulation to support your answers.

1 Why don't we get a lunar eclipse every single month?

2 How does your model show that the Moon doesn't produce its own light?

3 What would happen to the Moon's phases if the Moon orbited faster?

4 How are lunar eclipses and solar eclipses different in your model?

5 Why is gravity essential for the patterns we observe in the sky?

8 STEM Challenge: Design a Moon Phase Predictor

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YOUR ENGINEERING MISSION

Design a physical or digital tool that can predict the Moon's phase and eclipse events for any date, using your understanding of orbital mechanics.

THE SCENARIO

A local planetarium needs a hands-on exhibit that helps visitors understand why the Moon changes shape and when the next eclipse will occur. Your team has been hired to build the prototype.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO REPRESENT EARTH, MOON, AND SUN?
2. HOW WILL YOU SHOW THE MOON'S ORBITAL TILT AND WHY ECLIPSES ARE RARE?
3. HOW WILL YOUR EXHIBIT DEMONSTRATE WHY THE MOON TURNS RED DURING ECLIPSES?
4. HOW CAN VISITORS INTERACT WITH YOUR MODEL TO EXPLORE DIFFERENT PHASES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Planetary Scientists and Astronomers study celestial mechanics and predict eclipses, tides, and orbital events. They work at NASA, observatories, and universities, earning $70,000–$130,000/year.